## Summary

The interaction between biotic and abiotic components in an environment is called ecosystem. Autotrophs and heterotrophs are the producers and consumers respectively. The function of ecosystem refers to creation of energy, flow of energy and cycling of nutrients. The amount of light available for photosynthesis is called Photo synthetically Active Radiation . It is essential for increase in the productivity of ecosystem. The rate of biomass production per unit area /time is called productivity. It is classified as primary productivity, secondary productivity and community productivity. The transfer of energy in an ecosystem can be termed as energy flow. It is explained through the food chain, food web , ecological pyramids ( pyramid of number, biomass and energy ) and biogeochemical cycle. Cycling of nutrients between abiotic and biotic components is evident in the pond ecosystem, making itself self sufficient and self regulating Ecosystem protected for the welfare of posterity is called ecosystem management.

Successive replacement of one type of plant community by the other of the same area/ place is known as plant succession. The first invaded plants in a barren (nude) area are called pioneers (pioneers communities). On the other hand, a series of transitional developments of plant communities one after another in a given area are called seral communities. Succession is classified as primary succession, secondary succession, allogeneic succession and autotrophic succession. Plant succession is classified in to hydrosere (Initiating on a water bodies) ,Mesosere and xerosere. Further xerosere is subdivided in to Lithosere ( Initiating on a barren rock), Halosere and Pasmmosere.

Vegetation refers to the plant cover of an area. Geographically, India and Tamil Nadu show tropical climate. Hence it has rich vegetation (Forest vegetation, Grassland vegetation, Riparian vegetation, Aquatic and semi aquatic vegetation). According to Champion and Seth (1968), forest vegetation of India and Tamil Nadu has been classified in to 16 and 9 types respectively.

## Evaluation

I Choose the most suitable answer from the given four alternatives and write the option code and the corresponding answer.

1. Which of the following is not a abiotic component of the ecosystem?
   a) Bacteria
   b) Humus
   c) Organic compounds
   d) Inorganic compounds

2. Which of the following is / are not a natural ecosystem?
   a) Forest ecosystem
   b) Rice field
   c) Grassland ecosystem
   d) Desert ecosystem

3. Pond is a type of
   a) forest ecosystem
   b) grassland ecosystem
   c) marine ecosystem
   d) fresh water ecosystem

4. Pond ecosystem is
   a) not self sufficient and self regulating
   b) partially self sufficient and self regulating
   c) self sufficient and not self regulating
   d) self sufficient and self regulating

5. Profundal zone is predominated by heterotrophs in a pond ecosystem, because of
   a) with effective light penetration
   b) no effective light penetration
   c) complete absence of light
   d) a and b

6. Solar energy used by green plants for photosynthesis is only
   a) 2 – 8%  b) 2 – 10%



TN_GOVT_BOTANY_XII_PAGES_149-177 CH 07.indd 174 23-12-2022 11:03:07